Lapi architecture handover

Prepared for Swamy. This handover describes Lapi architecture, OCI resources, model placement, running versions and document workflows. Resource and model-mapping facts were inspected read-only on 15 October 2026 at 22:26 IST. Service health and source readiness were checked again on 16 October. Use Lapi_OCI_Deployment_Guide.pdf for the single maintained deployment procedure.

Lapi is running on Oracle Cloud Infrastructure in Mumbai. The web application, API, agents, worker and CPU reranker run on OKE. Oracle Autonomous AI Database 26ai stores application records, the durable job queue and native vector knowledge. OCI Object Storage holds the documents and saved result files.

The three inspected active datasets use Google Gemini for embeddings, chat, extraction and OCR. These model calls leave the OKE backend for the configured Google API. Laptop Ollama remains a separately registered alternative. It was stopped during the 15 October resource inspection; the later 16 October local check found its container healthy with zero restarts. Container health alone does not verify the complete cloud-to-laptop relay or a model inference.

Runs where	Components currently used
OCI OKE	Next.js web app, Caddy HTTPS, FastAPI, LangGraph agents, durable worker and CPU reranker
OCI Autonomous AI Database	Application records, access, durable job queue and Oracle native vector knowledge
OCI Object Storage	Original documents, prompts, parsed content and saved JSON / Parquet results
Google cloud API	Gemini embeddings, chat, extraction and dataset-aware OCR for all three active datasets
Local laptop	Retained Ollama / Qwen / Nomic option; Ollama healthy on 16 October; full relay/inference not exercised

The public HTTPS application returned 200. Both application deployments had one ready replica, and all five application containers were ready with zero restarts. The later read-only health check confirmed the same live images, pod identities and restart counts. No release, configuration change, document upload or model inference was performed. The manager branch is lapi_sprint1; the immutable original source freeze is lapi-freeze-2026-10-15.

Recently finalized Mural architecture

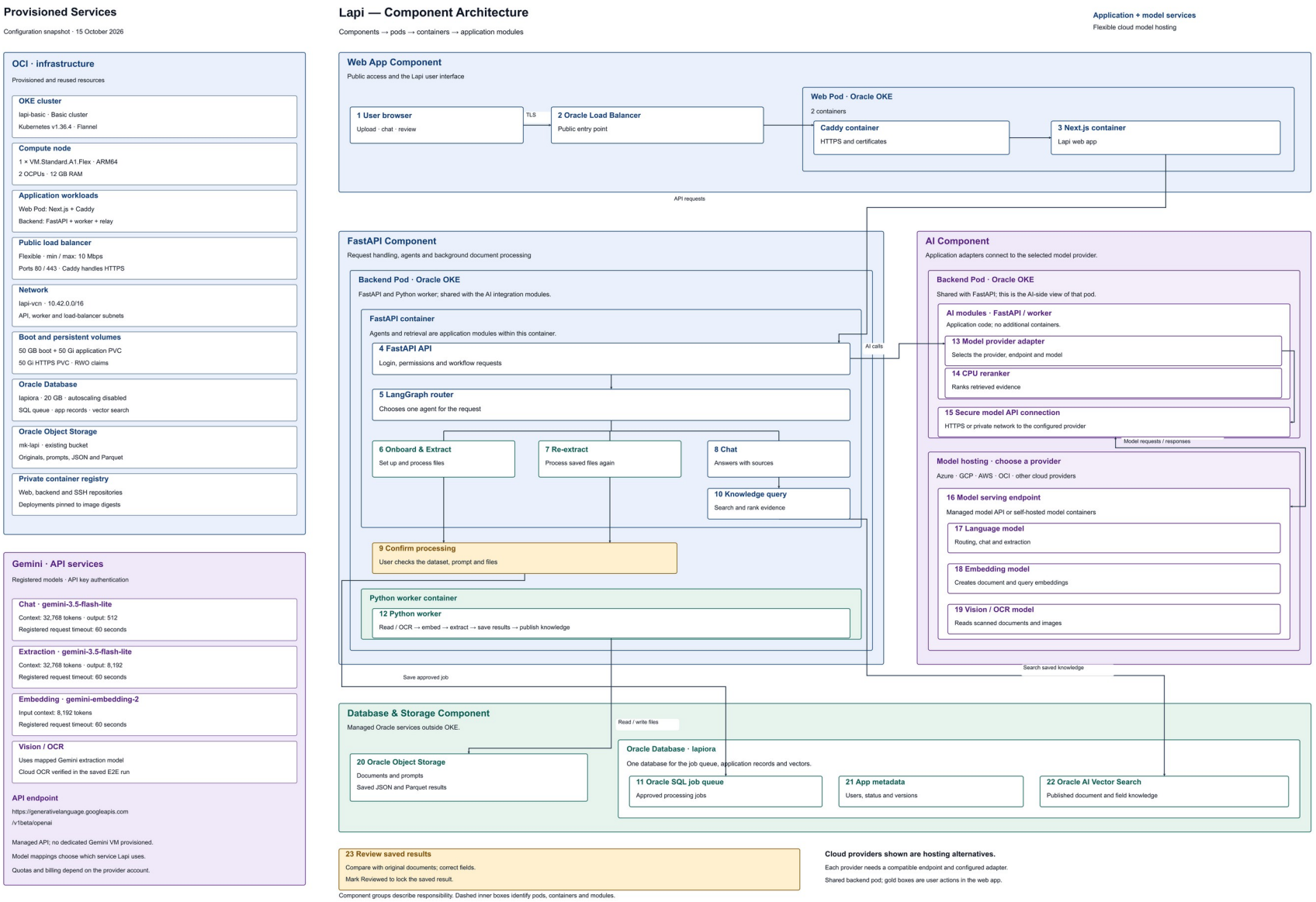


Figure 1 The existing Mural component diagram finalized on 15 October 2026, reused at full resolution with its corrected text spacing. Gemini is the active model path; other providers in the diagram are alternatives. Zoom in for component labels.

Verified OCI resources

These values were read from OCI and Kubernetes during the inspection. The database is the managed Autonomous AI Database service, not the stopped Oracle Free container on the laptop. Its live database version is 26ai and its compute allocation is 1 ECPU.

The two persistent claims each report 50 Gi in Kubernetes and are backed by OCI block volumes reporting 50 GB each. The worker also has a 50 GB boot volume, for 150 GB of allocated boot and block volumes. Database storage is a separate 20 GB allocation.

OCI resource	Live configuration	Observed state
Region and tenancy	Mumbai ap-mumbai-1; maniatramco	Authenticated access
OKE cluster	lapi-basic; Basic; Kubernetes v1.36.4	ACTIVE
Node pool	lapi-a1-node-pool; 1 node	ACTIVE
Compute	VM.Standard.A1.Flex; ARM64; 2 OCPUs; 12 GB RAM	RUNNING / Ready
Autonomous AI Database	lapi_ora_db / lapiora; Oracle 26ai; 1 ECPU; 20 GB	AVAILABLE
Database configuration	Free Tier flag true; compute/storage autoscaling off; mTLS required	Read connection verified
Object Storage	mk-lapi; Standard; namespace bmyljaecivrj	Read configuration verified
Public load balancer	92.4.67.183; flexible 10 Mbps minimum and maximum	ACTIVE
Application state	application-state; 50 Gi claim; 50 GB OCI volume	Bound / AVAILABLE
HTTPS state	https-state; 50 Gi claim; 50 GB OCI volume	Bound / AVAILABLE
Node boot volume	50 GB; 10 VPUs per GB	AVAILABLE
Private image registry	lapi-web, lapi-backend, lapi-ssh	All private

The bucket currently reports ObjectRead public access and versioning Disabled. Those existing settings were observed and left unchanged. The database Free Tier flag does not establish that the complete OKE environment has zero cost.

Running containers and storage connections

Deployment lapi-web contains web and https. Deployment lapi-api-worker contains api, worker and model-relay. Each deployment has one desired replica and one ready replica. Both use Recreate for updates.

The table shows configured resource requests and limits. A CPU value of 250m means one quarter of a core. These values are not current utilization; the cluster Metrics API was unavailable during the inspection.

Container	CPU and memory request	CPU and memory limit
api	250m CPU; 768Mi	1 CPU; 2Gi
worker	500m CPU; 2Gi	1500m CPU; 4Gi
model-relay	25m CPU; 32Mi	100m CPU; 128Mi
web	150m CPU; 512Mi	1 CPU; 1536Mi
https	50m CPU; 64Mi	250m CPU; 256Mi

1 Database connection. The active destination is OCI database lapiora through wallet alias lapiora_low, using mTLS. The configured search provider is oracle and the queue provider is database.

2 Content connection. The active content destination is OCI Object Storage bucket mk-lapi in namespace bmyljaecivrj. Originals, parsed content, prompts and versioned results keep their saved storage routes.

3 Persistent application state. API and worker share /data on application-state. It holds protected configuration, routes, locks, archives, model files and bounded application cache. Wallet and OCI credential mounts are separate.

4 HTTPS state. Caddy uses the separate https-state claim for persistent certificate state. Only the public entry is exposed for normal user access; FastAPI is an internal ClusterIP service on port 8000.

One node and one replica per deployment describe the current installation. A future application rollout can interrupt service; no rollout was performed for this documentation update.

Current AI models and local connections

All three active datasets map their Embedding, Chat and Extraction defaults to the Gemini registrations below. Dataset-aware OCR also resolves to the mapped Gemini extraction registration. Global Ollama defaults remain in configuration, but they do not override these saved dataset mappings.

Role	Mapped model	Where it executes
Embedding	gemini-embedding-2	Google Gemini API
Chat	gemini-3.5-flash-lite	Google Gemini API
Extraction	gemini-3.5-flash-lite	Google Gemini API
OCR for mapped datasets	gemini-3.5-flash-lite	Google Gemini API
Candidate reranking	Configured CPU reranker	OCI API / worker; model files on shared state

1 Gemini connection. The saved endpoint host is generativelanguage.googleapis.com. Embedding uses gemini-embedding-2; chat and extraction use gemini-3.5-flash-lite.

2 Saved model limits. The embedding registration has an 8192-token context limit. Chat has 32768 context tokens and 512 output tokens; extraction has 32768 context tokens and 8192 output tokens. These are saved application settings, not a vendor capacity claim.

3 Reranker location. The configured reranker is enabled in local CPU mode inside the OCI backend, with files at /data/models/reranker. Embeddings and generated answers still use the mapped Gemini API.

4 Retained laptop option. Qwen3 chat/extraction and Nomic embeddings remain enabled registrations. The older laptop application/database/index containers are separate from the OCI installation. Local Ollama was healthy in the 16 October check. Provider inference and full cloud relay connectivity were not exercised; use this route only when its saved mapping and relay are available.

5 OCR behavior. For these Gemini-mapped datasets, OCR uses the saved Gemini extraction model. The retained Qwen vision setting belongs to the local alternative path; it is not the current dataset OCR model.

This check read registrations and mappings without issuing a new Gemini inference request or restarting local models. Model availability and provider quota were not tested by making a paid processing call.

Developed document and answer workflows

1 Workspace and access. Administrators manage users, passwords, roles and dataset assignments. The API checks the signed-in user and selected dataset/document scope before access.

2 Dataset and prompt setup. A dataset maps its model roles. Prompt templates define the fields and validation rules for extraction. Template versions remain associated with saved results.

3 Confirmed document upload. The user confirms the target dataset, documents and required external-processing consent. Original bytes are saved in OCI Object Storage and durable jobs are recorded in Autonomous AI Database.

4 Parsing and indexing. The OCI worker reads the originals, parses text and uses the mapped Gemini model for OCR when needed. Gemini embeddings are published into Oracle native vector knowledge with document and page references.

5 Structured extraction. The worker retrieves and reranks evidence, calls Gemini with the saved prompt, validates the structured response, then saves versioned JSON and Parquet results and publishes result knowledge.

6 Review and export. Users compare saved fields with original documents, correct values and mark a result Reviewed to save and lock it. Exports use saved data. Re-extract creates another result while preserving earlier versions.

7 Unified Chat. Every question follows access and scope checks, hybrid vector and keyword retrieval, CPU reranking, Gemini answer generation, and answer/citation validation. Lapi help follows the same retrieval path. Exact counts use complete authorized catalog facts.

8 Recovery. Accepted worker jobs persist beyond the browser. Progress comes from saved job state. Retries recover the accepted workflow and knowledge publication rather than silently submitting duplicate processing.

The inspection verified configuration and service readiness. It did not run a new upload, extraction, chat question, account change or recovery action.

Versions observed in running services

The application versions below were read from the running containers. Kubernetes and Oracle Database versions were read from live OCI and node information. This page intentionally focuses on deployed technologies and omits package and library dependency inventories.

Running technology	Observed version	Location
Oracle Autonomous AI Database	26ai	OCI managed database
OKE Kubernetes	v1.36.4	OCI Mumbai
Worker operating system	Oracle Linux Server 9.8	OCI ARM64 node
Node.js	22.23.3	OCI web container
Next.js	15.5.27	OCI web container
React	19.3.0	OCI web container
Python	3.12.15	OCI API and worker image
FastAPI	0.142.2	OCI API container
LangGraph	1.2.14	OCI backend agent runtime
Caddy	2.10.2	OCI HTTPS sidecar

The currently mapped model identifiers are gemini-embedding-2 and gemini-3.5-flash-lite. Cloud model identifiers are saved provider selections rather than installed weight versions. Full deployed image references are recorded in the final section of this document.

Network and operating boundaries

The VCN is lapi-vcn with CIDR 10.42.0.0/16. Its API, worker and load-balancer subnets are 10.42.0.0/24, 10.42.1.0/24 and 10.42.2.0/24. The load balancer exposes ports 80 and 443 at 92.4.67.183. The application uses HTTPS and returned the HSTS header.

The internal API service uses port 8000. The retained Ollama service uses port 11434, and the restricted SSH relay service uses NodePort 32022. The relay sidecar was ready at inspection. The later local Ollama health result does not prove end-to-end relay connectivity. No model port or network policy was changed.

The database is available, mTLS is required, and both database compute and storage autoscaling are disabled. The database records the queue and native vectors in the same managed service. The bucket is Standard storage; original documents and complete result files remain in Object Storage.

OCI API calls were GET/list operations. Kubernetes reads inspected nodes, deployments, pods, services and persistent claims; read-only commands inside existing containers checked versions and configuration. The application catalog query explicitly used a read-only database transaction and bypassed cache writes.

The inspected installation has a single worker node and one replica per deployment. The backend release strategy is Recreate, so an application rollout can briefly interrupt the API and worker. Do not promise uninterrupted releases or independent scaling without redesigning shared persistent state.

For a daily pause, first finish chat and all queued or running processing. An application replica pause leaves the worker VM allocated. Stopping the existing Compute VM through OCI Console stops both application pods and preserves the disks; start that same VM and wait for application readiness. Stopped compute does not remove storage, registry, load balancer or database allocations.

Docker Desktop, laptop Ollama and the SSH tunnel are needed only for a deliberately selected laptop model mapping. The three inspected active datasets use Gemini and do not require that tunnel. Do not apply the separate local Compose stack to the existing cloud workspace.

Read-only diagnostics: kubectl get deployments,pods, then inspect events and the relevant container logs using the explicit destination kubeconfig, context and namespace. Keep support output containing private application data outside Git.

Maintain the application and approved design

The application contracts are recorded in root AGENTS.md and the release skill at .agents/skills/lapi-release/SKILL.md. Every Unified Chat question must enforce access and scope, retrieve using semantic and keyword evidence, rerank, generate and validate the answer and citations. Exact counts require complete authorized facts. Do not change the processing agents to implement a chat-only feature.

Use lapi_frontend/app/workflow-stepper.tsx and its shared stylesheet for real progress. Completed stages are blue with white checks; an active stage uses a navy rotating ring and particles only on its incoming completed connector. Pending stages remain static. Labels and times describe actual server events; never fabricate percentages, stages or completion. Preserve failure states, keyboard inspection, responsive fitting and reduced-motion behavior.

Use lapi_frontend/app/agent-logo.tsx for Agent identity and preserve the existing platform mark in the header. Earlier design reference images remain in Git history. The recently finalized Mural diagram is embedded in this document and requires no external assets.

Runtime values come from lapi_backend/lapi_app/lapi_config/runtime.toml; saved model registrations can impose smaller limits. Keep credentials, bootstrap tokens, wallets and deployment receipts out of Git. Source-declared versions and observed container versions are separate facts.

Source area	Responsibility
lapi_backend/lapi_app	FastAPI, access, agents, worker, provider and storage adapters
lapi_frontend/app	Next.js interface, API proxy, shared progress and identity
lapi_database	Oracle SQL definitions and migration boundaries
lapi_deploy	OCI release, authentication and runtime tooling
lapi_local_use	Local runtime profiles, launchers, Oracle Free setup and optional Kubernetes
lapi_testing	Backend/frontend tests, fixtures, QA scripts and test containers

For a separate local installation, use lapi_local_use/lapi_scripts/setup.sh and lapi_local_use/compose.yaml, or the Windows Oracle launcher lapi_local_use/lapi_scripts/start-oracle.ps1. Use independent data and secrets. These development alternatives do not describe the live OCI topology. Root README links this architecture reference and the single final OCI deployment PDF.

Source freeze and validation evidence

The final handover branch is lapi_sprint1, created from the python source freeze. Tag lapi-freeze-2026-10-15 remains unchanged. The tag identifies source, not proof that the deployed images are byte-for-byte builds of that commit. The source-only manifest at the tag fingerprints 481 inputs; later documentation and folder changes on lapi_sprint1 do not redeploy the application.

Recorded check	Actual result
Full backend regression	16 October 2026: 1047 passed, 5 skipped; temporary SQLite and mock providers
Test folder reorganization	Tests, fixtures and QA tools are under lapi_testing; imports, CI, Compose and container paths verified
Frontend unit suite	167 passed; both type checks and ARM64 production build/startup passed; browser workflows not executed
OCI runtime profile	Live-matching Python/Node bases and all 105 backend package versions; model configuration unchanged
Deployment tooling	69 offline deployment tests passed; scripts and skill validated; no application rollout performed
Live inspection	HTTPS 200, both replicas ready, five containers ready with zero restarts; no inference, uploads or writes
Document consolidation	The recent Mural image is reused; Word pages are rendered and visually checked; links and source boundaries are verified

The original freeze used local Python 3.12.8 and Node.js 24.19.0. The later release-readiness check built ARM64 images with the live-matching Python 3.12.15 and Node.js 22.23.3 bases; all 105 installed backend package versions matched live. It passed 1,047 backend tests with 5 skipped, 167 frontend unit tests, frontend type checks and 69 offline deployment tests. These are source checks, not a live release.

GitHub Actions triggers on main pushes and pull requests. A direct push to lapi_sprint1 is not a fresh CI run and does not deploy the application. The verified application/tooling baseline is 3ae03a5f40fa24bd7c05f84892a9d73efddc677d. No new-tenancy installation, live provider inference, upload/extraction/chat write test or recovery drill was performed.

Superseded deployment guides and earlier handover versions are archived under delete/ with their original paths. Lapi_OCI_Deployment_Guide.pdf is the current deployment authority for both the AI skill and JSON plus PowerShell routes. Resource observations retain their inspection timestamps; no live settings were changed for this handover.

Live image and diagram provenance

The following image references were recorded from the live deployments during the 15 October 2026 inspection. They identify that observation and were not pushed or changed for this documentation work.

```
Web container
bom.ocir.io/bmyljaecivrj/lapi-web
sha256:12336119afcca52b7eb9bbc34329cc967279787398406a637b52d2b4740b31fc

API and worker containers
bom.ocir.io/bmyljaecivrj/lapi-backend
sha256:fe2ea263dd76ef2a1b50a56b2b26cab0eb81da43cbf0c852dd270f5b8d046ae5
Running platform image ID
sha256:25fc43cab4ba8be40549383b803753277d6043a003a84193803182c35096a780

Model relay sidecar
bom.ocir.io/bmyljaecivrj/lapi-ssh
sha256:37189e82533c3e893f261b9735880e74f521f2ee6ce15f4490e3087ea1944870

HTTPS sidecar
docker.io/library/caddy
sha256:4c6e91c6ed0e2fa03efd5b44747b625fec79bc9cd06ac5235a779726618e530d
```

The backend desired image is an index digest; the running image ID is its platform digest. Both are recorded separately. The diagram is the corrected full-resolution image finalized on 15 October and placed on the existing Mural board; its model-hosting section shows supported alternatives while the live mapping is Gemini.

Mural board: https://app.mural.co/t/clouddataai5577/m/clouddataai5577/1789398606376/9cd26abe7fa9fe9b3ae2ba45aaa4a4722ad240f0?wid=0-1792066520255

Original source manifest: https://github.com/Maniatramco/Lapi/blob/lapi-freeze-2026-10-15/lapi_docs/RELEASE-MANIFEST.json